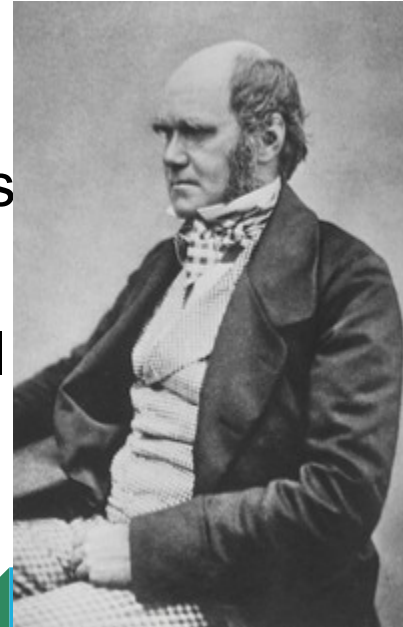


Adaptation Through Natural Selection - Changes in Environmental Conditions and Extinctions (47-51)



Adaptation Through Natural Selection

- Charles Darwin
 - theory of evolution through natural selection
 - Journey on HMS Beagle[found diverse species and extinct species]
- **1859:**“The Origin of Species by Means of Natural Selection”
 - Organisms reproduce diverse offspring
 - Most offspring will die, while some survive to reproduce



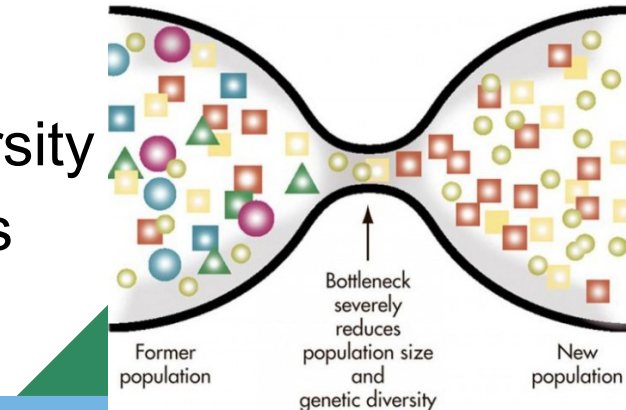
Adapting to a Changing Environment

- Animals cannot respond to too big of a change
 - Either die out or adapt
- Adaptability of a species to change depends on how much, and how fast, change happens
 - Birds eating nuts
 1. Slowly harder nuts => birds have strong beaks
 2. Suddenly all super hard nuts => birds can crack nuts



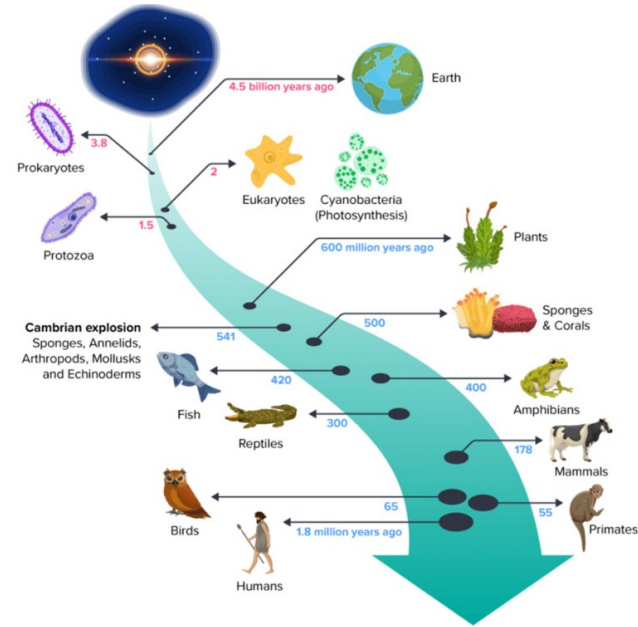
Nonadaptive Evolutionary Processes

- **Gene flow**: different populations of the same species mix
- **Genetic Drift**: mutation in genotypes among small populations of a species
- low genetic variation is correlated to increased disease risk and low fertility
 - Cheetah populations and genetic diversity dropped so much that every cheetah is identical



The Pace of Evolution

- an adaptation can take centuries or millennia
- 3 things affect pace of adaptation
 - Rate of change in environment
 - Genetic variation
 - Population size
 - Smaller populations undergo nonadaptive mechanisms



Thank

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You!